

E - algebraic operations with exponentials, logarithms etc.

$$(a) \frac{2}{\sqrt{2}} = \frac{\sqrt{2} \times \sqrt{2}}{\sqrt{2}} = 2^{\frac{1}{2}}$$

$$(b) \sqrt[4]{8} = \sqrt[4]{2^3} = 2^{\frac{3}{4}}$$

$$(c) 2 \cdot 4^{2x} = 2 \cdot (2^2)^{2x} = 2^{4x+1}$$

$$(d) \frac{2^{x+y}}{2^{x-y}} = 2^{x-x+y-(-y)} = 2^{2y}$$

$$(e) (2^x)^x = 2^{x^2}$$

$$(f) \frac{2^5 \cdot 2^{\frac{1}{5}} \cdot 2^{\frac{3}{5}}}{2^{-\frac{1}{5}}} = 2^{5+\frac{1}{5}+\frac{3}{5}+\frac{1}{5}} = 2^{6+1} = 2^7$$

$$2(a) 3a^4 b^3 \cdot 2a^3 b^{-2} = 6a^7 b^1$$

$$(b) 3x^{2n+3-(n-1)} = 3^{n+4}$$

$$(d) 3^{(x+1)(x+1)} = 3^{x^2+2x+1}$$

$$(c) 3^{-4} x^{2x-4} y^{-3x-4} z^0 = 3^{-4} x^{2x-4} y^{-3x-4} = \frac{y^2}{81x^3}$$

$$(e) x^{\frac{1}{3}(1-y)} = x^{\frac{1}{3}-\frac{y}{3}} = x^{\frac{1-y}{3}}$$

$$(f) \sqrt{9x^2} = 3|x| = 3x \text{ because } \begin{cases} x \geq 0 \\ -x < 0 \end{cases}$$

$$(g) 4a^{\frac{1}{2}-2x-\frac{1}{2}} b^{6x-\frac{1}{2}} = \frac{a}{2b^3}$$

$$(h) \frac{(x^{\frac{1}{2}})^{-\frac{1}{2}}}{(x^{-\frac{1}{2}})^{-\frac{1}{2}}} = \frac{x^{-\frac{1}{4}}}{x^{\frac{1}{4}}} = \frac{1}{x^{\frac{1}{2}}}$$

$$(i) \frac{7^{-1} x^{\frac{1}{4}}}{3^{-2} y^{-1}} \times \frac{(49y)^0}{2(3^{-\frac{1}{2}})^2} = \frac{7^2 x^{\frac{1}{4}} y}{7} \times \frac{1}{2(3^{-1})} = \frac{3^2 x^{\frac{1}{4}} y}{14} = \frac{27}{14} \sqrt[4]{x} y$$

$$(j) \frac{2^{(n-1)(n-1)}}{2^{(n+1)(n+1)}} = 2^{n^2-2n+1-(n^2+2n+1)} = 2^{-4n}$$

$$(k) \frac{(2)^{-\frac{3}{4}} (a)^{-\frac{3}{4}}}{(3)^{-\frac{3}{4}} (a^{-1})^{-\frac{3}{4}}} = \frac{4^{-\frac{3}{4}} a^{-\frac{9}{4}}}{3^{-\frac{3}{4}} a^{\frac{3}{4}}} = \frac{(2^2)^{-\frac{3}{4}} \times 3^3 \times a^{-\frac{9}{4}-\frac{3}{4}}}{1} = \frac{1}{2^3} \times 3^3 \times a^{-3} = \frac{27}{80^3}$$

$$(l) \frac{(3 \times 4)^n \times (2^3)^{n-1} \times 3^{n+1}}{2 \times 3 \times 4 \times (2 \times 3)^{n-2} \times 2^n} = \frac{3^n \times 2^{2n} \times 2^{3n-3} \times 3^{n+1}}{2 \times 3 \times 2^2 \times 2^{n-2} \times 3^{n-2} \times 2^n}$$

$$= \frac{2^{5n-3} \times 3^{2n+1}}{2^{3+n-2+n} \times 3^{(2n+1)-(n-1)}} = \frac{2^{5n-3-(2n+1)} \times 3^{(2n+1)-(n-1)}}{1} = 2^{3n-4} \times 3^{n+2}$$

$$(m) \sqrt[3]{a} \cdot \sqrt{a} = a^{\frac{1}{3}} \cdot a^{\frac{1}{2}} = a^{\frac{1}{3}+\frac{1}{2}} = a^{\frac{2+3}{6}} = a^{\frac{5}{6}} = \sqrt[6]{a^5}$$

$$(n) 3^{n-n^2} \times (2 \times 3)^{n+1} \times \frac{2^{-1} \times (3^2)^{-n-1}}{3^{(1-n)(n+1)}} = 3^{n-n^2+1+n+1} \times 2^1 \times 3^{-2n-2+(n^2-1)} = 2^0 \times 3^{-n^2+2n+2+n^2-2n-3} = 3^{-1} = \frac{1}{3}$$

$$(o) \frac{x^{8n-3} x^{-6n} (y^{\frac{1}{2}})^{-6n} x^3}{x^{-4n} y^{-4n} x^{-n}} = \frac{x^{8n-3-6n+3+4n+n} y^{-3n+4n}}{1} = x^{7n} y^n$$

$$(p) \frac{(3^2)^n - 3^{2n+1}}{3^{2n} - 3^{n+2+n}} = \frac{3^{2n} - 3^{2n} \cdot 3}{3^{2n} - 3^{2n} \cdot 3^2} = \frac{3^{2n}(1-3)}{3^{2n}(1-9)} = \frac{-2}{-8} = \frac{1}{4}$$

$$(q) \frac{5^x - 3 \times 5^x \times 5^{-2}}{5^x \times 5 + 3 \times 5^x} = \frac{5^x(1-3^2 \times 5^{-2})}{5^x(5+3)} = \left(1-\frac{9}{25}\right) \times \frac{1}{8} = \frac{16}{25} \times \frac{1}{8} = \frac{2}{25}$$

$$(r) (a^{\frac{1}{2}} - b^{\frac{3}{2}})^2 = (a^{\frac{1}{2}} - b^{\frac{3}{2}})(a^{\frac{1}{2}} - b^{\frac{3}{2}}) = a^1 - 2a^{\frac{1}{2}}b^{\frac{3}{2}} + b^{\frac{9}{2}} = a - 2\sqrt{ab^3} + b^3$$

$$\begin{aligned} (5) \quad \frac{a^{-1}}{x^{-1}-a^{-1}} &= \frac{\frac{1}{a}}{\frac{1}{x}-\frac{1}{a}} \\ &= \frac{\frac{1}{a}}{\frac{a-x}{ax}} = \frac{1}{a} \div \frac{a-x}{ax} \\ &= \frac{1}{a} \times \frac{ax}{a-x} \\ &= \frac{x}{a-x} \end{aligned}$$

$$\begin{aligned} (t) \quad \left(\frac{(3y)^{-2}}{3y^{-2}} \right)^{\frac{2}{3}} &= \left(\frac{3^{-2}y^{-2}}{3y^{-2}} \right)^{\frac{2}{3}} \\ &= \frac{3^{-\frac{4}{3}}y^{-\frac{4}{3}}}{3^{\frac{2}{3}}y^{-\frac{4}{3}}} = \frac{y^{\frac{4}{3}}}{3^{\frac{2}{3}}3^{\frac{4}{3}}y^{\frac{4}{3}}} \\ &= \frac{1}{3^{\frac{6}{3}}} = \frac{1}{3^2} = \frac{1}{9} \end{aligned}$$

$$\begin{aligned} 3. a) \quad \log \left(\frac{xy}{z} \right) \\ &= \log x + \log y - \log z \end{aligned}$$

$$\begin{aligned} b) \quad \log \sqrt{\frac{x}{y}} &= \log \left(\frac{x}{y} \right)^{\frac{1}{2}} \\ &= \frac{1}{2} (\log x - \log y) \end{aligned}$$

$$\begin{aligned} c) \quad \log(x^2y^3) &= \log x^2 + \log y^3 \\ &= 2\log x + 3\log y \end{aligned}$$

$$\begin{aligned} d) \quad \log \left(\frac{1}{x^2y} \right) &= \log 1 - \log x^2y \\ &= \log 1 - (2\log x + \log y) \\ &= -2\log x - \log y \end{aligned}$$

$$\begin{aligned} e) \quad \ln \left(\frac{xy^2}{x+y} \right) &= \ln(xy^2) - \ln(x+y) \\ &= \ln x + 2\ln y - \ln(x+y) \end{aligned}$$

$$\begin{aligned} f) \quad \ln \sqrt[3]{x^2+1} &= \ln(x^2+1)^{\frac{1}{3}} \\ &= \frac{1}{3} \ln(x^2+1) \end{aligned}$$

$$\begin{aligned} g) \quad \ln \sqrt[3]{x^2y} &= \ln(x^2y)^{\frac{1}{3}} \\ &= \frac{1}{3} [2\ln x + \ln y] \\ &= \frac{2}{3} \ln x + \frac{1}{3} \ln y \end{aligned}$$

$$\begin{aligned} (2) \quad h) \quad \ln \sqrt{\frac{x^3}{y}} &= \ln \left(\frac{x^3}{y} \right)^{\frac{1}{2}} \\ &= \frac{1}{2} (3\ln x - \ln y) \\ &= \frac{3}{2} \ln x - \frac{1}{2} \ln y \end{aligned}$$

$$(i) \quad \log \left(\frac{x}{yz} \right) = \log x - \log y - \log z$$

$$\begin{aligned} (j) \quad \ln \left(\frac{(x+y)^3}{xy^2} \right) \\ &= 3\ln(x+y) - (\ln x + 2\ln y) \\ &= 3\ln(x+y) - \ln x - 2\ln y \end{aligned}$$

$$(4) \quad a) \quad 3\ln x = \ln x^3$$

$$b) \quad 3\ln x - \frac{1}{2}\ln y = \ln \left(\frac{x^3}{\sqrt{y}} \right)$$

$$\begin{aligned} c) \quad \frac{1}{2}(\ln x - 3\ln y) &= \frac{1}{2} \ln \left(\frac{x}{y^3} \right) \\ &= \ln \sqrt{\frac{x}{y^3}} \end{aligned}$$

$$\begin{aligned} d) \quad 2\ln x + \ln(x+y) \\ &= \ln[(x^2)(x+y)] \end{aligned}$$

$$\begin{aligned} e) \quad 2\ln x - 3\ln y + \ln z \\ &= \ln \left(\frac{x^2z}{y^3} \right) \end{aligned}$$

$$\begin{aligned} (f) \quad \ln(y-z) + \ln y - 2\ln x \\ &= \ln \left(\frac{y(y-z)}{x^2} \right) \end{aligned}$$

$$\begin{aligned} g) \quad \frac{1}{2} [\ln x - 5(\ln y - 3\ln z)] \\ &= \ln \left(\frac{xz^{15}}{y^5} \right)^{\frac{1}{2}} = \ln \sqrt{\frac{xz^{15}}{y^5}} \end{aligned}$$

$$\begin{aligned} h) \quad \frac{1}{2} [(\log x - 5\log y) - 3\log z] \\ &= \frac{1}{2} [\log x - 5\log y - 3\log z] \\ &= \log \sqrt{\frac{x}{y^5z^3}} \end{aligned}$$

$$(i) \quad \log \left(\frac{5x^3}{16y} \right)$$

$$(j) \quad \log \left(\frac{x^3y^3}{x^2y} \right) = \log(xy^2)$$

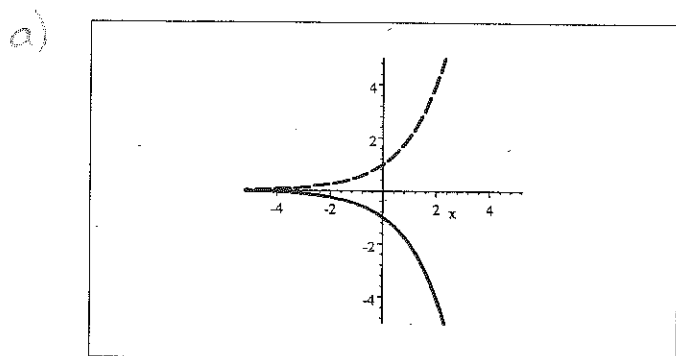
$$\textcircled{5} \text{ a) } \log_a 27 = \log_a 3^3 \\ = 3 \log_a 3 = 3y$$

$$\text{b) } \log_a \frac{2}{9} = \log_a 2 - \log_a 3^2 \\ = x - 2y$$

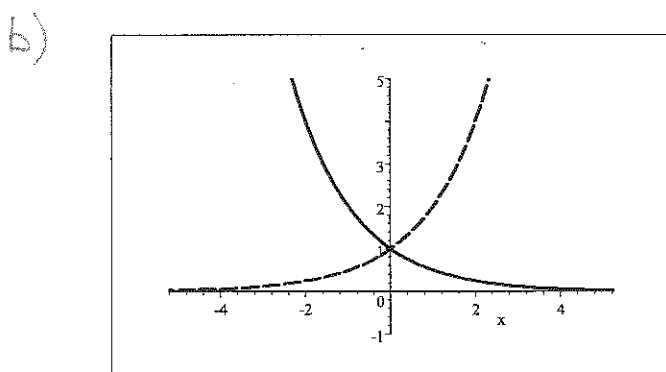
$$\text{c) } \log_a 6^{1/3} = \frac{1}{3} \log_a (2 \times 3) \\ = \frac{1}{3} (\log_a 2 + \log_a 3) \\ = \frac{1}{3} (x + y)$$

$$\text{d) } \log_a \sqrt{12a} = \frac{1}{2} [\log_a 12 + \log_a a] \\ = \frac{1}{2} [\log_a (4 \times 3) + 1] \\ = \frac{1}{2} [\log_a 2^2 + \log_a 3 + 1] \\ = \frac{1}{2} [2 \log_a 2 + \log_a 3 + 1] \\ = x + \frac{y}{2} + \frac{1}{2}$$

$$\textcircled{6} y = f(x) = 2^x$$

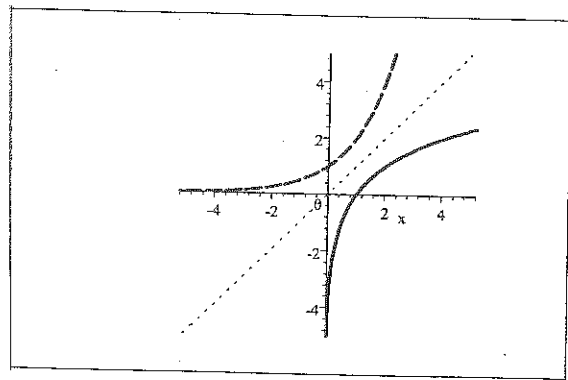


[g —, f ---]



[h —, f ---]

③ c)



[k —, f ---, y=x]

$$7) \log_a b = 3 \Rightarrow a^3 = b \\ \Rightarrow a = b^{1/3} \Rightarrow \log_b a = 1/3$$

$$8) y = \frac{r}{1 + ce^{-at}} \\ \Rightarrow y(1 + ce^{-at}) = r \\ \Rightarrow y + yce^{-at} = r \\ \Rightarrow yce^{-at} = r - y \\ \Rightarrow e^{-at} = \frac{r - y}{cy} \\ \Rightarrow -at = \ln\left(\frac{r - y}{cy}\right)$$

$$9) f(x) = y = \log_b x \\ \bullet f\left(\frac{1}{b}\right) = \log_b \frac{1}{b} = \log_b b^{-1} \\ = -1 \log_b b = -1 \\ \bullet f(\sqrt{b}) = \log_b b^{1/2} = \frac{1}{2} \log_b b \\ = \frac{1}{2}$$

$$10) \log_b \left(\frac{1}{10}\right) = -\frac{1}{2} \\ \Rightarrow -\log_b 10 = -\frac{1}{2} \\ \Rightarrow \log_b 10 = \frac{1}{2} \\ \Rightarrow b^{1/2} = 10 \\ \Rightarrow b = 10^2 = 100$$

$$\textcircled{11} \text{ a) } 4(e^x)^4 + (x^e)^4 \\ = 4e^{4x} + x^{4e}$$

$$\text{b) } \frac{1}{2} e^{2x} \cdot 4e^{\frac{3}{2}} = 2e^{2x + \frac{3}{2}} \\ = 2e^{\frac{2x^2 + 3}{x}}, x > 0$$

$$\text{c) } \frac{14e^{3x}}{7e^x} = 2e^{2x}$$

$$\text{d) } \sqrt{e^x} \cdot \sqrt[3]{e^{-3x}} = e^{\frac{x}{2}} \cdot e^{-\frac{x}{3}} \\ = e^{\frac{x}{2} - \frac{x}{3}} = e^{-\frac{x}{6}}$$

$$\text{e) } \ln(e^{3x}) = 3x \ln e = 3x$$

$$\text{f) } e^{\ln 7x} = 7x, x > 0$$

$$\text{g) } \ln(9e^x \cdot 10e^{2x}) \\ = \ln(90e^{3x}) \\ = \ln 90 + \ln e^{3x} \\ = \ln 90 + 3x, x \in \mathbb{R}$$

$$\text{h) } e^{\ln(\frac{x^2}{y})} = \frac{x^2}{y} \text{ where } \frac{x^2}{y} > 0 \\ \text{and } y \neq 0$$

$$\text{i) } 3\ln x + \ln \frac{1}{x} \\ = 3\ln x - \ln x = 2\ln x, x > 0$$

$$\text{j) } \ln e^2 + e^{-\ln x} \\ = 2\ln e + e^{\ln \frac{1}{x}} \\ = 2 + \frac{1}{x} \\ = \frac{2x+1}{x}, x > 0 \text{ and } x \neq 0$$

$$\textcircled{12} \text{ a) } \ln e^{4x} = 12 \\ \Rightarrow 4x \ln e = 12 \\ \Rightarrow x = \frac{12}{4} = 3$$

$$\text{b) } e^{4x} = 12 \\ \Rightarrow \ln e^{4x} = \ln 12 \\ \Rightarrow 4x = \ln 12 \\ \Rightarrow x = \frac{\ln 12}{4}$$

$$\textcircled{4} \text{ c) } \ln 4x = 12 \\ \Rightarrow e^{\ln 4x} = e^{12} \\ \Rightarrow 4x = e^{12} \Rightarrow x = \frac{e^{12}}{4}$$

$$\text{d) } e^{2\ln 3} = x \\ \Rightarrow e^{\ln 3^2} = x \Rightarrow x = 3^2 = 9$$

$$\text{e) } \ln x = 3\ln 2 - 2\ln 3 \\ \Rightarrow \ln x = \ln 2^3 - \ln 3^2 \\ = \ln \frac{8}{9} \\ \Rightarrow x = \frac{8}{9}$$

$$\text{f) } \frac{1}{2} \ln x = 1 - \ln 2 \\ \Rightarrow \ln x^{\frac{1}{2}} = \ln e - \ln 2 \\ = \ln \left(\frac{e}{2}\right) \\ \Rightarrow x^{\frac{1}{2}} = \frac{e}{2} \Rightarrow x = \left(\frac{e}{2}\right)^2 \\ \Rightarrow x = \frac{e^2}{4}$$